Internetworking over asynchronous transfer mode networks

T J King, G M Whalley, P Brown and L N C Morgan

The creation of wide area data networks between multiple customer sites, known as internetworking, is a key growth area for BT. Customer networks are constructed using complex data networking devices known as 'routers'. These are combined with one or more network services such as BT's switched multi-megabit data service (SMDS), frame relay, ISDN, flexible bandwidth service (FBS) and private leased circuits to form customer networks of a required performance and cost. The emerging asynchronous transfer mode (ATM) technology brings both interesting new possibilities and challenges to future internetworks. This paper describes the current status of ATM data networking in both the wide area and local area network environments (WAN/ LAN), then discusses the key challenges to network designers, network operators and equipment vendors.

1. Introduction

Internetworking is the creation of wide area data networks, that allow local area networks (LANs) and computers to be connected. It includes the extension of customer data protocols across the wide area, how routes across the network are determined and how end-stations are located.

BT already runs and manages many such internetworks, both on behalf of our customers and for our internal use. End systems connected include mainframes, front end processors (FEP), workstations and desktop computers. BT has an excellent portfolio of wide area data services including the switched multi-megabit data service (SMDS), frame relay (FR), the integrated services digital network (ISDN), the flexible bandwidth service (FBS) and several LAN extension services that are 'mixed and matched' to provide internetworking solutions to meet customer performance and cost requirements. A simplified topology is shown in Fig 1, where typically two of the underlying network services are combined to produce the customer's internetwork.

BT is trialling a relatively new technology, known as asynchronous transfer mode (ATM), that has the potential to provide integrated transport and switching for data, video and voice services. ATM could therefore be used as another underlying technology to create internetworks for customers, with the potential benefits of increased throughput, lower end-to-end network latency (delay) and integration with non-data services. Additionally, there is much activity in the area of ATM LANs which is currently subject

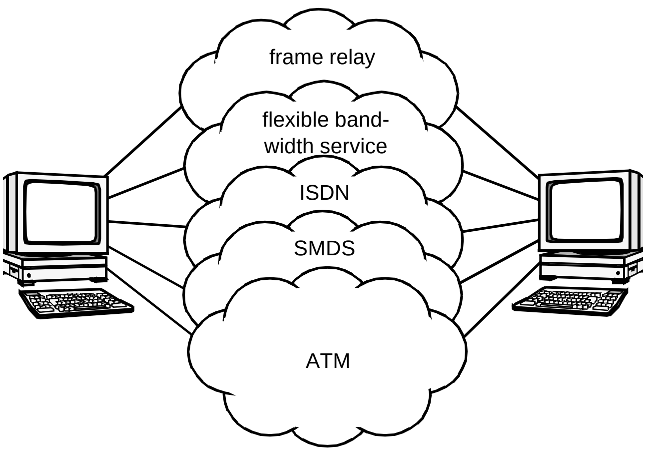


Fig 1 The internetworking concept.

to a more detailed study at BT Laboratories. With 'ATM-to-the-desktop' becoming a rapidly expanding market, internetworking of ATM LANs is a new opportunity for service providers. This paper focuses on the use of ATM for data networks and the issues that are being or need to be addressed so that scalable networks can reliably be produced using the technology. It starts by outlining BT's 'data over ATM' trials, the options for ATM in the LAN and the WAN, then discusses in detail how such internetworks are being designed.

2. ATM in wide area and local area networks (WAN/LAN)

There are already several ATM trials in place worldwide, both in the LAN and WAN, not to mention the first ATM service offerings in Europe and the USA. BT are studying the use of ATM in both the WAN and LAN scenarios, and are involved in trialling the technology in both these areas. The most important of these trials for BT is the European ATM pilot, which links 18 operators in 16 European countries. The European pilot is particularly examining the effects of supporting existing data services on an ATM core. Examples of the data services being supported on the ATM pilot are shown in Fig 2.

With the advent of ATM, equipment vendors have responded quickly to produce, and in some cases modify existing, customer premises equipment (CPE) which allow customers to access ATM networks. These solutions leave the customers' existing LANs intact, whilst utilising the standards-defined ATM adaptation layers (AAL) to translate user traffic from the high layers of the protocol stack into a size and format that can be contained within the payload of an ATM cell. Each AAL consists of two sublayers — the segmentation and reassembly (SAR) sublayer and the convergence sublayer (CS). There are currently two AALs in use for data services over ATM — AAL3/4 and AAL5. AAL3/4 is designed to handle data traffic that can tolerate delay, but not cell loss, and to accomplish this AAL3/4 performs error detection and, if required, correction on each cell. However, AAL3/4 is regarded by some as being sophisticated and inefficient (the AAL3/4 overhead consumes 4 bytes of the 48-byte ATM cell payload). AAL5 handles data traffic with less overhead than AAL3/4, hence its original title of simple and efficient adaptation layer (SEAL).

There are two ways in which ATM will become the technology to support customers' data networks. Firstly, ATM will provide the WAN infrastructure to support BT's existing data services for customers, e.g. SMDS and frame relay. This solution has the benefits of a single technology core to support multi-megabit link rates, and opens the possibilities for integrated network management functions. Secondly, ATM will be used in the LAN, both for high-performance workgroups (ATM to the desktop), and as the backbone for the support of legacy protocols (e.g. LAN emulation). These scenarios are described in more detail in the following paragraphs.

2.1 *Data network access to the ATM WAN*

Currently, ATM equipment vendors are enabling access to ATM WANs through either upgrades to their existing CPE equipment, primarily routers, or the introduction of new products which are essentially interworking units that provide address mapping (e.g. E.164 or DLCI) to the appropriate VPI/VCI (Fig 3). The data service unit (DSU) illustrated could even be an interface card on an ATM switch. The list of ATM CPE vendors is ever increasing, and includes not only the established telecommunications

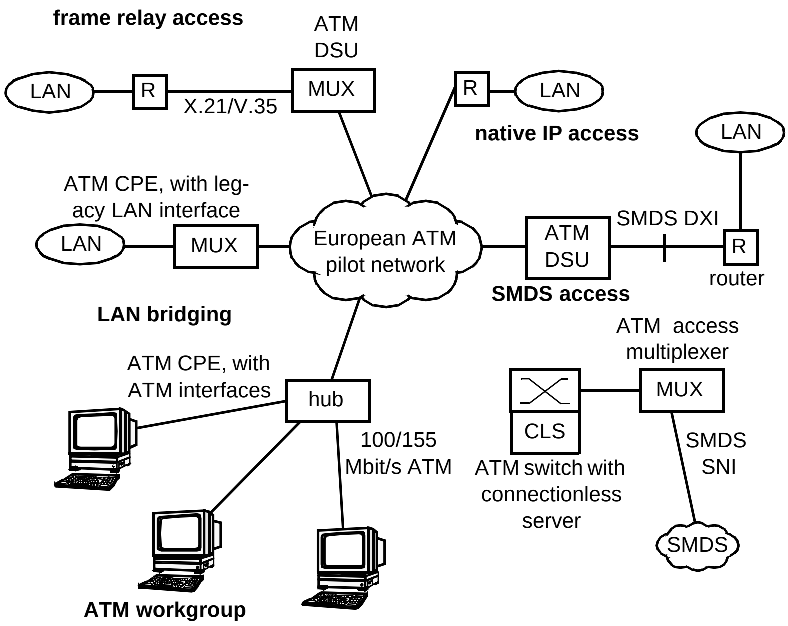
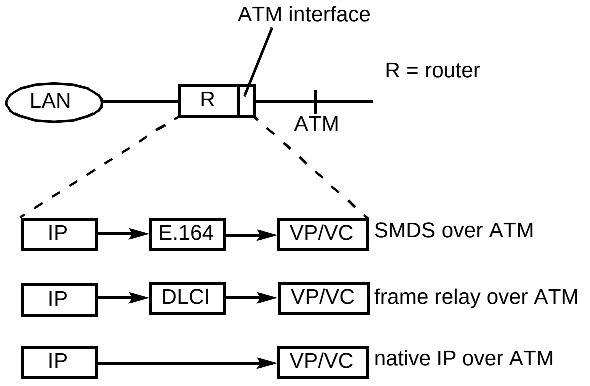
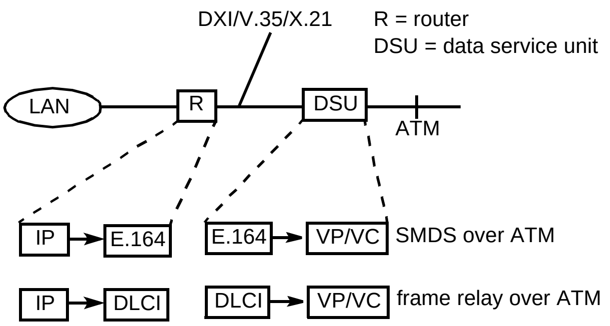


Fig 2 Data services on the European ATM pilot.



(a) Router with integrated ATM interface.



(b) DSU providing ATM access.

Fig 3 Access solutions for the ATM WAN.

manufacturers, but those previously associated with computer networking equipment, and new start-up ventures.

2.2 ATM in the LAN and ATM LAN emulation

ATM to the desktop is already a reality, with ATM cards available for workstations, at 100 and 155 Mbit/s, enabling high-power workgroups to be realised. However, most of the data traffic in existing customer premises networks is sent over LANs such as IEEE 802.3 (Ethernet) and IEEE 802.5 (Token Ring). Thus, no matter how great the benefits of ATM prove to be, there is going to be a very large installed base of legacy LANs that need to be catered for, for the foreseeable future. Several ATM CPE vendors have made a first pass at this market-place by bridging LAN traffic over a full ATM mesh, which is impractical for a large network over the wide area. To address this issue, the ATM Forum have produced their LAN emulation specification [1], which defines how Ethernet or Token-ring connected devices can communicate with their counterparts over an ATM network. The aim of LAN emulation is to make the ATM network invisible to the legacy LANs, making it possible to benefit from the performance enhancements of ATM without the expense of hardware or software changes to the end devices. Also, LAN emulation allows legacy LANs to communicate with ATM-based end devices which are equipped with the LAN emulation software (Fig 4).

The emulated LAN has two main components:

- LAN emulation clients (LECs),
- the LAN emulation service, which is implemented on three logical servers, the LAN emulation server (LES), the broadcast and unknown server (BUS), and the configuration server (note that these three functions could be a single system, or they could be distributed throughout the ATM network, in the LAN or WAN).

The LEC software resides either on the ATM-connected end-system, or on an interworking unit between the ATM network and the legacy LAN. The LES controls the LAN emulation service, and is responsible for resolving MAC addresses to ATM addresses. The BUS handles all broadcasting and multicasting, and the configuration server typically holds information on the emulated LAN which a LEC is about to join. When the MAC address has been resolved to the ATM address, the user data can be transferred using either an SVC or a PVC. However, if LAN emulation is to avoid becoming totally unmanageable, SVCs must be used. ATM Forum UNI 3.0 SVC capability is already available on the smaller ATM switches which are typically found on a customer's premises, but SVCs are unlikely to be available in the WAN until near the end of this decade.

ATM LAN emulation offers a number of advantages over legacy LANs, for example a LEC can be located anywhere on the ATM network, i.e. it does not suffer from the distance limitation imposed on conventional LANs. Congestion is much less on an emulated LAN, as most of the traffic is carried on dedicated connections per end user. One of the unresolved issues in LAN emulation, however, is that the protocol for interconnecting LESs has not yet been finalised and, at present, this is vendor-specific. It should be noted that LAN emulation only provides for communication between like-media, e.g. emulated Ethernet-to-Ethernet; Ethernet-to-Token ring would require the use of a translational bridge/router. As LAN emulation is a Layer 2 function, it is independent of the protocol being routed, e.g. TCP/IP or IPX.

To complete the ATM LAN story, mention must be made of the Internet Engineering Task Force's (IETF) 'Classical IP over ATM' specification, as defined in RFC 1577 [2]. In contrast with LAN emulation which supports all protocols, Classical IP only supports IP, although, because it does not emulate the existing MAC layer, the amount of overhead is less. It is designed to take advantage of ATM features such as specifying cell delay, QoS, and the

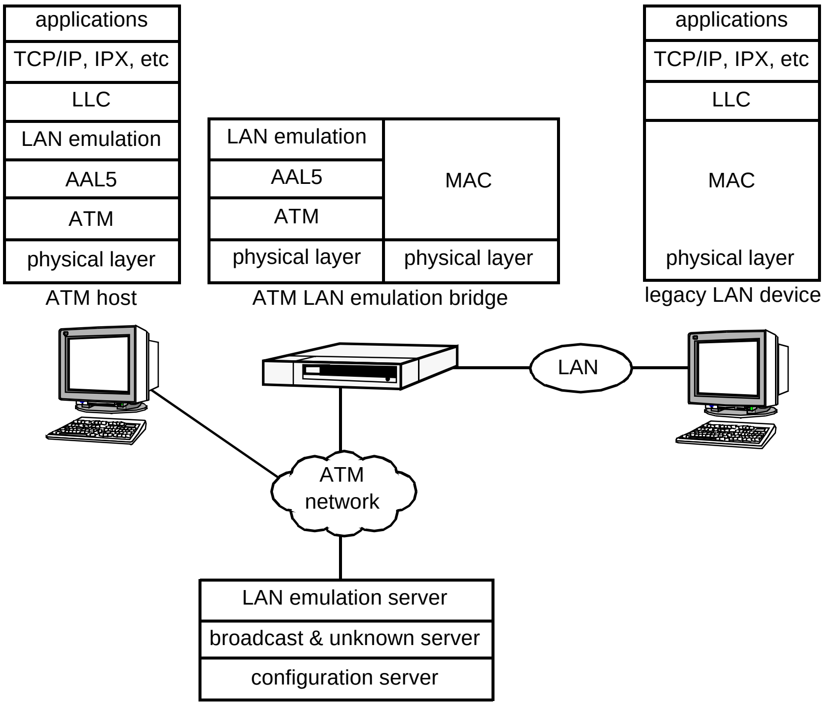


Fig 4 ATM forum LAN emulation.

maximum PDU size, which can exceed 9000 bytes. Interworking with devices connected to legacy LAN systems is only possible via a router which is running Classical IP over ATM.

3. The impact of scale

Scalability in today's internetworking solutions can add significant complexity to the logical network design (e.g. routeing/addressing strategy to achieve the required connectivity), particularly where enhanced resilience or support for multiple protocols is a requirement. A number of factors influence where this added complexity comes from, but probably the single most important point is that the network layer routeing protocols for most of the protocol suites in common use today were not designed for 'large cloud networks'. Examples of large cloud networks are SMDS and ATM where hundreds, potentially thousands of multiprotocol routers are likely to be connected to the same multi-access network, only a single hop away from each other.

Considering the example in Fig 5, which is representative of the current approach to large scale router network design using SMDS, the customer virtual private network (VPN) is broken down into a number of smaller groups, with each of these including at least two routers acting as 'gateways' between the groups. The effect of this is to give 'any to any' connectivity within a group with two or three hops for communication between the groups.

The equivalent model for a large scale ATM-based internetwork is shown in Fig 6, which treats the customer VPN as a single group or cloud.

It is likely that IP, in native form or to transport other protocols through encapsulation, will have a significant role to play in early and future ATM-based internetworks. The following sections compare the two approaches assuming TCP/IP as the protocol.

3.1 Routeing

The overhead traffic associated with dynamic routeing protocols is proportional to the number of routers on the same multi-access network. Using multiple smaller groups allows the bandwidth available for customer traffic to be maximised and also reduces the router processing overhead for the dynamic routeing process. Treating the network as a single large cloud per customer, there is a need for a new breed of routeing protocols as discussed later.

Failures in the network will cause the dynamic routeing protocol to update all routeing tables with the topology change and where possible decide upon an alternative route. As well as rendering the routes unusable for the time it takes

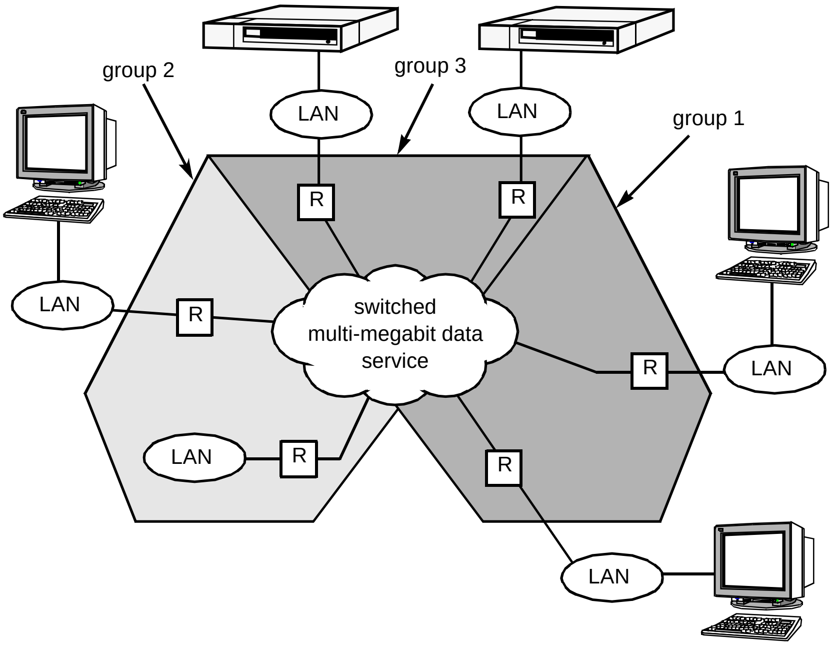


Fig 5 Model for large scale SMDS design.

the routers to ‘converge’ on a new topology, significant extra traffic is generated during this period of uncertainty. By restricting the number of routers per group, failures can be resolved locally, adding stability to the overall design and reducing the disruption to the rest of the network. While stability implications need careful consideration for a single large routeing domain, at the same time this must not affect the convergence performance of today’s protocols.

Many of today’s routeing protocols are not optimised to work on large cloud networks and grew out of the traditional wide area network environment which predominantly used private leased circuits. Typically, they have more emphasis on fast convergence and have performance rather than policy based metrics. The model in Fig 5 is required for these protocols to work.

3.2 Addressing

Sites with low-speed access (less than 2 Mbit/s) would typically have a ‘low end’ router, which would, though,

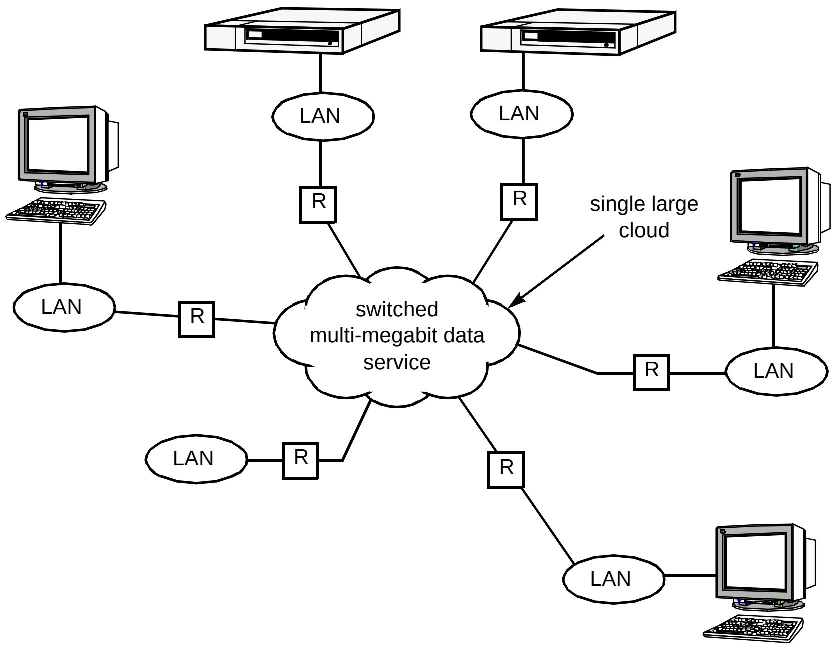


Fig 6 Large scale ATM internetworking.

have more than enough processing power to saturate a 2 Mbit/s link, but would not have the ability to process and store routes to, say, 1000 networks. Protocol suites such as TCP/IP which have hierarchical addressing schemes can allow many routes to be summarised by a single address, where the single address is reachable through a gateway router which has more explicit information. The way in which the address space is broken down to achieve this summarisation is key and must be consistent across the whole internetwork. The method for providing a similar address summarisation function on a large cloud network is less clear though is equally important where 'any to any' connectivity is required.

3.3 Protocol support

Most customers have at least three protocols which they want to run across the wide area. The most common examples are TCP/IP, IBM protocols and Novell Netware. TCP/IP and associated open shortest path first (OSPF) routeing protocol supports the model in Fig 5. IBM SNA traffic has a proprietary network layer which cannot be routed by normal means; encapsulation in TCP/IP is the only viable option currently for large-scale multi-protocol networks and thus SNA traffic could share the same stable IP routeing core as native TCP/IP traffic. Novell Netware using IPX presents a greater problem in that it does not support hierarchical routeing at present and, using the model in Fig 5, care would need to be taken to keep overhead traffic levels manageable in a large network. Most applications running on these protocols have a need for fast convergence under failure, this is particularly true for session-based protocols using SNA.

ATM will help indirectly with the above issues, primarily because it will act as a catalyst for the development of a new generation of routeing and address resolution protocols which are optimised for large cloud networks. This is already under-way through the IETF work on routeing over large clouds (ROLC).

3.4 The need for improved routeing protocols

To overcome many of the issues discussed so far with respect to routeing protocols and to a lesser extent for addressing, new protocols are required which are optimised for large cloud networks. Such protocols would need to exhibit the following characteristics:

- low overhead traffic even when scaled to hundreds/ thousands of routers,
- single hop connectivity,
- stability under failure conditions,
- flexibility to define a preferred path through the network,
- fast convergence to cope with, for example, IBM protocols,
- low processing overhead.

Such a protocol does not currently exist, though much work is under-way in the ROLC group to look at address resolution protocols, which is closely related to the routeing process but solving a different problem. The next hop resolution protocol (NHRP) [3], in principle, can be used to complement a very limited subset of existing routeing protocols (e.g. the border gateway protocol, BGP4) for IP in an environment where multiple 'logical IP subnetworks' (LIS) exist within a single customer VPN. A simplified version of this is currently running successfully with the UK academic community where, though logically the network is subdivided into domains, the BGP protocol allows the path of the routeing traffic (multi-hop) and the path of the data traffic (single hop) to be separated. BT Laboratories were involved in implementing this on SMDS. NHRP offers an intra- and inter-LIS resolution protocol so that, in theory, where a destination is in a different LIS, the routeing and address resolution can give the network and link (e.g. E164 address for SMDS, VP/VC for ATM) address of the next hop rather than the 'gateway' router described earlier in this section. NHRP is also applicable to end systems and thus it could be argued that there is no need for a network layer routeing protocol at all. The route server model could work in conjunction with NHRP to give single hop connectivity to end systems running NHRP; however, in the absence of SVCs, large networks would be difficult to administer and dynamic resilience would not be possible.

4. Multicasting requirements

The role of multicasting for addressing and routeing functions is a powerful tool in the local area network environment. A similar facility is advantageous across the wide area network to simplify logical internetwork design. The ability to deliver a control packet to many destinations at the same time allows for greater autonomy through dynamic discovery of networks and nodes connected to those networks. As an example, if a router knows about three networks which are directly attached to it and wishes to advertise this to one hundred routers in the same domain attached to an ATM network, then, without multicast, each router needs to have 99 neighbours statically defined so that a copy of the routeing packet containing reachability of the three networks can be copied to each of the 99 neighbours. With multicast, each router simply needs to be configured with the same multicast identifier and a single copy of the routeing packet will be delivered to all 99 neighbours.

Figure 7 shows multiple routers attached to an ATM network. The network shown has evolved from the existing technology platform for SMDS (DQDB 802.6) to one using ATM. The customer still has exactly the same SMDS

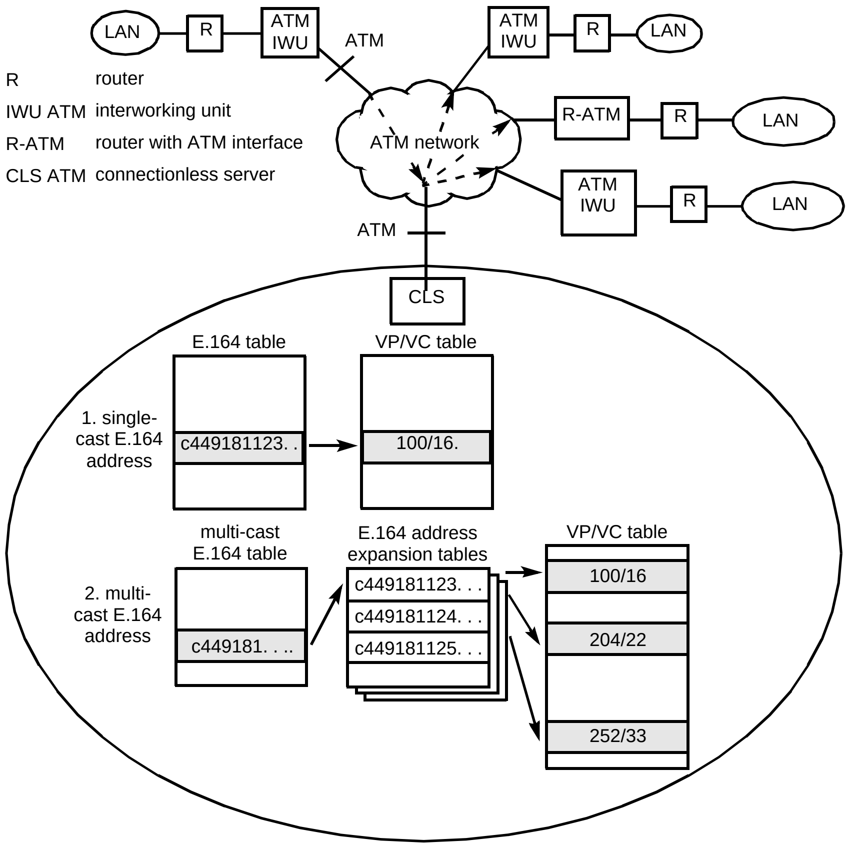


Fig 7 ATM connectionless server operation.

service, but now on an ATM technology platform. This brings with it issues concerning SMDS E.164 address to ATM VP/VC translation and support for the multicasting feature. Early ATM switch implementations make use of connectionless servers (CLS) to achieve these functions.

Considering a single-cast packet first, this would be sent from the customer end station to the router which would add SMDS encapsulation. This would then be encapsulated in AAL3/4 with the VP/VC of the connectionless server. As shown in Fig 7, the connectionless server would use the E.164 destination address to index the relevant VP/VC of the destination and the packet forwarded appropriately. A multicast packet uses a similar principle, except this time there is an added stage with the multicast E.164 address used to index the E.164 address expansion table, which derives the E.164 individual addresses of all members of the group. The individual addresses are then applied to the same VP/VC table for the single-cast case and the VP/VC of each member of the group derived. The E.164 destination address for each of the resultant packets would be the original multicast E.164 address to maintain transparency. Thus individual E.164 addresses are only used in the translation process.

In large SMDS networks with several CLSs, group addressing must be carefully administered to prevent the group address resolution (or ‘explosion’) happening at the wrong point in the network. For example, a group addressed packet from the UK may have several members in Germany. It is sensible therefore to forward only one copy of the packet to the CLS in Germany, where the explosion can be performed. A major drawback with current CLS implementations is that the operator manually programs the E.164-to-VP/VC translations in the CLS, using PVCs. Should a link fail from the CLS, then there is no way for packets to reach their destinations without extensive re-routeing in the CLS and ATM network. Future CLS implementations may introduce SVCs, or possibly a dynamic routeing protocol between CLSs.

5. Resilience aspects

Resilience is crucial to any network, with downtime and network crashes causing loss of data and customer confidence. Clearly, a reliable and workable resilience strategy for ATM is essential.

ATM has the ability to multiplex data, voice and video into cell streams. The voice and video cell streams are very sensitive to delay during transmission and these types of data are often termed isochronous (sensitive to delay). Only the resilience for data will be described in this paper. A typical scenario of a primary and back-up network is shown in Fig 8.

Integrated services digital network (ISDN) is one of the most widely used network technologies for increasing resilience in a primary network. However, ISDN only works at relatively slow data transfer rates compared to ATM; for example, with a 34 Mbit/s ATM link, a 128 kbit/s basic-rate ISDN will only provide 0.5% of the original bandwidth, and primary-rate ISDN is only a little better at 8%. Frame relay operates at similar speeds to ISDN. Clearly, ISDN or frame relay could only be used for very slow ATM links, or where a minimum bandwidth at this level is acceptable in the rare occurrence of failure.

For ATM with speeds up to 34 Mbit/s, SMDS could be used to provide resilience for data on the primary ATM links. SMDS is an well-established technology and well-suited to the transfer of data where transmission delays are less critical. When the required ATM access bandwidth is much higher, typically 155 Mbit/s and all data services need to be retained, ATM itself is the only technology that could be used as a back-up. This could be provided by dual access from the customer's premises to the ATM network.

The ability to dynamically utilise back-up networks lies in the network layer routeing protocol, further information can be found on routeing protocol in the Routeing section of this paper.

6. Address resolution considerations

Address resolution provides the mapping between network layer addresses and physical addresses. There are two approaches — direct (static) mapping and dynamically learnt maps. For the specific network layer IP, dynamic learning is provided by the address resolution protocol (ARP). The protocol is defined generically so that it can be applied to a variety of underlying networks and protocols.

ARP allows us to assign network layer (IP in this example) addresses to machines entirely independently of the underlying physical hardware addresses. This can be particularly useful where customers' internetworks are created over public data networks such as ATM, since the IP addresses are often chosen by the customer, whereas the 'physical' addresses (the data equivalent of telephone numbers) are normally assigned by the carrier. ARP allows a machine to find dynamically the physical address of its destination, given only its IP address.

The less attractive alternative to such a dynamic binding algorithm is direct (or static) mapping. In this case, routers attached to the public network hold statically entered tables of destination IP and public-network physical address pairs. The creation and maintenance of such tables becomes a significant overhead in large internetworks. As an example, imagine a 500-site router network and assume for the purpose of this example that each router needs direct connectivity with each other. If the customer now wishes to add a 501st site, each of the existing 500 routers would each need a new address pair added to its static table.

The simplified operation of ARP is shown in Fig 9.

The mappings required in an ATM-based network are as shown in Figs 3 and 7. Currently these are manually entered into the router, DSU or CLS as appropriate. As networks scale and as required changes increase (e.g. a customer opens a new branch office), then the problem of maintaining these static mappings also increases. The requirement is for a dynamic means of creating and updating these mappings that is secure and reliable. A new form of resolution protocol is therefore required.

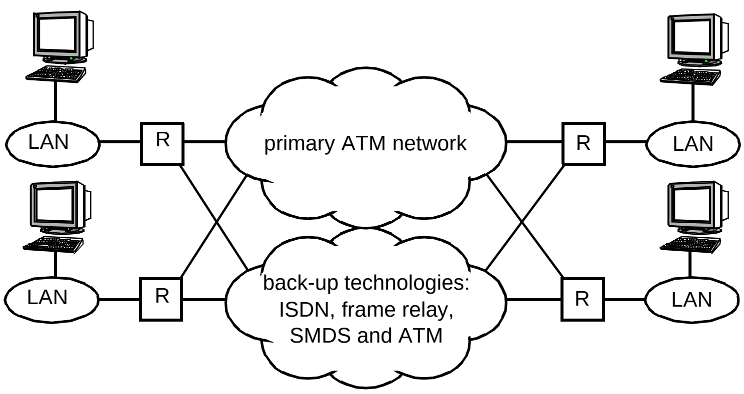


Fig 8 Example of a primary and back-up network.

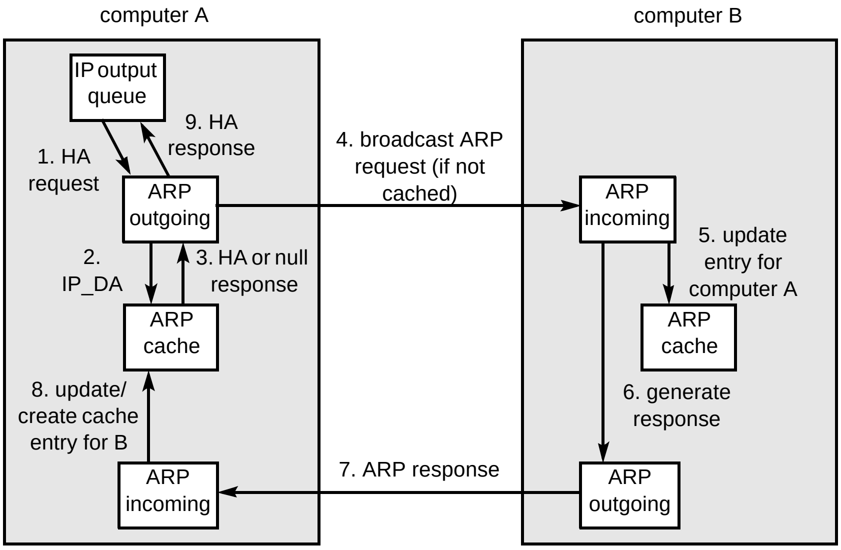


Fig 9 Simplified ARP operation.

7. Future development

The future for ATM is assured, primarily through the activities of the ATM Forum, that have allowed standards to be set to which equipment vendors have been able to produce. With the experience gained from the current trials and ATM service launches, customers and operators can now look forward to the introduction of SVCs and congestion control. These features will help to move ATM away from the current approach, which, although powerful in terms of bandwidth, is unwieldy with large numbers of customers as PVCs must be manually configured. Further refinement of the currently offered network management systems will also make ATM an attractive technology for internetworking. It is expected that these new features will be trialled in the LAN and WAN during 1995. Also the first LAN emulation products will be released by vendors, which together with the above features, could signal the widespread adoption of ATM as the preferred internetworking solution. In the near future, it is likely that ATM will find its way into BT's existing SMDS and frame relay networks, as a high-speed core technology. The introduction and its evolution thereafter will draw heavily on the experience gained through trials, such as the European ATM pilot.

ATM has much potential for internetworking in the future. The realisation of this potential can only be achieved if the dynamic routeing and address resolution functions can be supported on ATM networks, thus easing the operational support issues. The complexity of internetworking solutions being demanded by customers today, based on existing BT data services, must be matched and exceeded by any ATM-based solution. In practice, in the absence of switched virtual circuits and dynamic ATM layer routeing protocols, large static tables are inevitable and with the consequent maintenance overhead.

The TCP/IP protocol suite both in native form and to transport other protocols looks set to increase in the future. The key factors driving this are the agreement of the next generation of IP (IP Version 6) which offers significantly more address space than its predecessor (the current Version 4), plus the data link switching standard which defines a generic means of encapsulating the IBM protocols in TCP/IP that alleviates many of the shortcomings of the current approach.

BT has won many large contracts to provide hundreds and in some cases thousands of routers on frame relay and SMDS with ISDN back-up figuring in many cases to enhance resilience. The vast majority of customer accesses into the underlying services are 'low speed' (64K, 448K, 2M) using frame-based technology as shown in Fig 10. To offer a smooth migration to ATM, either the ATM interface needs to be after the first point of concentration, or frame-based ATM must be supported at these low speeds. The former approach is preferred in the first instance as it involves no changes in CPE hardware or software configuration. Where a customer needs more bandwidth for data applications, a direct ATM interface from his CPE may be the preferred option.

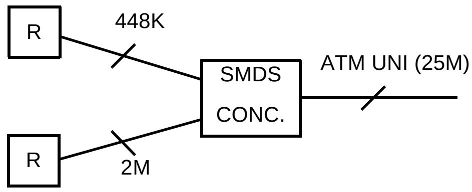


Fig 10 Evolution of low-speed access.

8. Conclusions

The paper has provided a picture of internetworking over ATM. One of the main conclusions is that the static mapping approach currently used in public ATM trials between network-layer and physical-layer addresses does not scale well and that there is a clear need for dynamic binding mechanisms. Secondly, large-scale networks will benefit from advanced routeing protocols that keep inter-router traffic at sensible levels. Thirdly, that market forces are currently deciding which of the various ways of providing internetworks over ATM will predominate.

The next few years are certain to prove very exciting as some of the more advanced concepts presented here are implemented in the equipment and rolled out as part of wide-area network services.

Acknowledgements

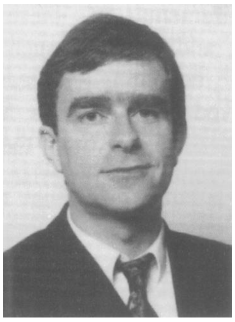
Many thanks to members of the internetworking and ATM teams at BT Laboratories and especially Robin Rowley and Adrian Smith, without whom this paper would not have been possible.

References

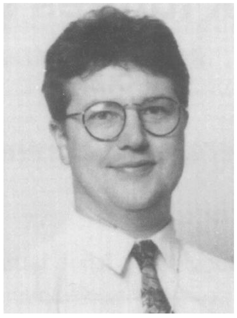
1 ATM Forum: 'LAN Emulation Specification' (1995).

2 Laubach M: 'Classical IP and ARP Over ATM', RFC 1577 (1994).

3 Katz D and Piscitello D: 'Next-hop resolution protocol (NHRP)', NBMA, Draft 4 (1995).

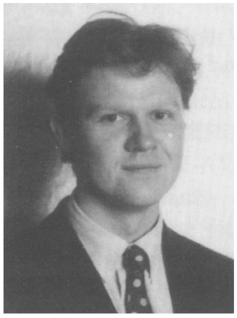


Tim King studied at Cambridge University during 1980—83 and holds an MA Hon degree. He joined BT Laboratories in 1988 and leads the internetworking team, working closely with end customers and account teams to provide world-class data internetworking. He was technically responsible for the trial and pilot phases of BT's switched multimegabit data service (SMDS). He has previously worked on ATM at BT Laboratories during 1988—1991, high-speed data networking for BICC Data Networks (now 3COM), including FDDI bridge product development and representing them on FDDI at international standards committees. He has also worked on broadband telecommunications, including ATM and high-speed data on cable TV systems, at the GEC Hirst Research Centre, UK. He is a Chartered Engineer and a Member of the IEEE and the IEE.



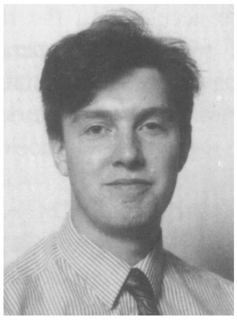
Glenn Whalley graduated in Communication Engineering from Leeds Polytechnic in 1988. He joined BT Laboratories in 1988 and is now a senior member of BT's internetworking team. He has worked on several aspects of ATM networks, and more recently high speed data networking across the wide area. He is a key member of the team that was responsible for the successful SMDS trial and pilot phases through to service launch in December 1993, particularly in the area of customer internetworking over SMDS.

Currently he is involved in the technical support of a number of customers and potential customers seeking leading edge internetworking solutions based on SMDS and other key BT service offerings.



Peter Brown joined BT Laboratories in 1991 having obtained a BSc in Electrical Engineering from Queen's University Belfast, and an MSc in Telecommunications from Essex University. He joined the satellite systems group and worked on data services over Inmarsat-B/M, before joining the ATM technology group in 1993. He was responsible for the testing and implementation of data services on the European ATM pilot, and was involved in a number of ATM demonstrations between the Broadband Laboratory at BTL and other European operations.

He is currently involved in the evaluation of a wide range of ATM equipment for possible use in the BT network.



Luke Morgan joined BT Laboratories in 1994, after graduating from the University of Strathclyde with a degree in Applied Physics in 1991 and studying for a PhD in Superconducting Physics, which he is still completing.

He joined the internetworking team in broadband multiservice networks, where he works extensively testing and investigating complex and large-scale internetworking solutions.